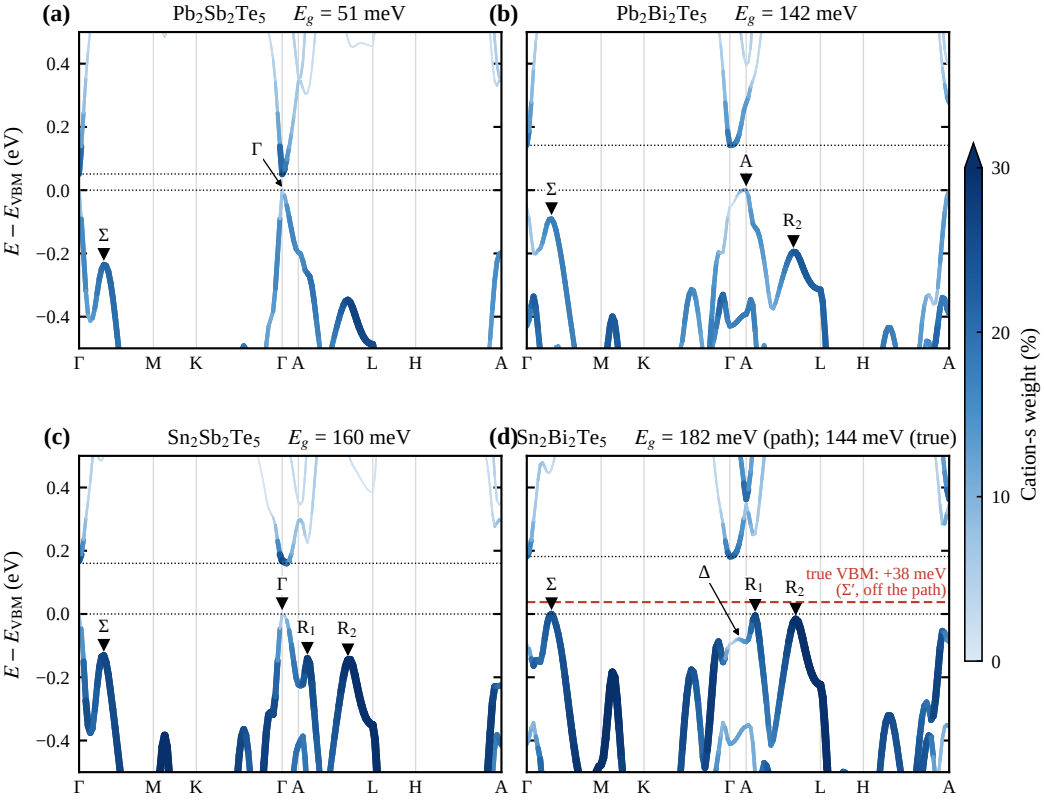

(a)
Pb2Sb2Te5
Eg = 51 meV
(b)
Pb2Bi2Te5
Eg = 142 meV
(c)
Sn2Sb2Te5
Eg = 160 meV
(d)
Sn2Bi2Te5
Eg = 182 meV (path); 144 meV (true)
E − EVBM (eV)
0.4
0.2
0.0
−0.2
−0.4
Γ
M
K
Γ A
L
H
A
Σ
Γ
A
R2
R1
Δ
true VBM: +38 meV
(Σ', off the path)
Cation-s weight (%)
30
20
10
0